

Run 07 | TEA

Global (including India)

IV + controls

Target Total early-stage entrepreneurial activity; binary 1 vs. 0
Sample 1,342,884 respondents **Coverage** 89 countries; 2015–2022
Positive cases 154,420 (11.50%) **Case time** 46m 07s
Train / val. / test 913,163 / 161,145 / 268,576 (68% / 12% / 20%, stratified)
Predictors (13): Self-efficacy, opportunity perception, fear of failure, knowing entrepreneurs, intention, media coverage, business exit, business angel, established business owner, good career choice, ease of starting a business, high status, social enterprise.
Fraser EFW predictors (7): Overall economic freedom, Economic freedom quartile, Size of government, Legal system and property rights, Sound money, Freedom to trade internationally, Regulation. Total independent variables: 20.
Controls (10): Survey year, Age, Gender, Education attainment, Household income, Household size, Occupational status, GDP per capita (PPP), Population, GDP growth. Country and respondent ID are excluded as predictors.
Protocol: Ten CPU models; model seed 20260925; fixed splits by sample/outcome; training-only preprocessing; F1 threshold selected on validation data. Test scores below are unweighted.

Result in brief

Highest observed ROC-AUC: Random Forest (0.9937). Highest F1: Random Forest (0.8875). For Random Forest, adding controls changed ROC-AUC by +0.0024 on the same test records.

Model performance on held-out test records

Model	ROC-AUC	PR-AUC	Acc.%	Prec.%	Recall%	F1	Time
Random Forest	0.9937	0.9527	97.35	86.77	90.83	0.8875	2m 38s
XGBoost	0.9936	0.9520	97.35	88.30	88.73	0.8851	22s
CatBoost	0.9933	0.9499	97.29	88.00	88.55	0.8827	3m 27s
H2O GBM	0.9921	0.9374	96.97	85.31	88.95	0.8709	39s
Lasso	0.9914	0.9327	96.89	84.92	88.67	0.8675	10m 39s
Elastic Net	0.9913	0.9327	96.89	84.95	88.64	0.8676	12m 34s
Logistic Regression	0.9912	0.9323	96.88	84.91	88.62	0.8672	1m 55s
Ridge	0.9909	0.9304	96.81	84.73	88.16	0.8641	8m 20s
Decision Tree	0.9892	0.8951	96.75	86.06	85.57	0.8581	13s
Naive Bayes	0.9240	0.5041	89.95	54.27	80.24	0.6475	4m 27s

Models ranked by ROC-AUC; small differences are descriptive, not significance tests. Time includes fit, prediction and output processing. Logistic regression uses a small stabilizing penalty; Lasso/Ridge/Elastic Net use 3-fold training CV.

Random-forest feature importance: top 8

Rank	Predictor	Raw importance	Share%
1	Entrepreneurial intention	71,776.80	46.07
2	Established business owner	29,199.75	18.74
3	Occupational status	11,851.08	7.61
4	Self-efficacy	3,978.49	2.55
5	Age	3,754.89	2.41
6	Knowing entrepreneurs	3,252.09	2.09
7	Size of government	2,561.12	1.64
8	GDP per capita (PPP)	2,383.07	1.53

Share is the percentage of total forest impurity importance across all 53 encoded predictors, not explained variance. Raw values are not comparable across samples. Missing indicators are derived preprocessing features.

Leading Fraser EFW predictors within the same forest

Rank	Fraser EFW predictor	Raw importance	Share%
1	Size of government	2,561.12	1.64
2	Legal system and property rights	2,038.13	1.31
3	Regulation	1,661.79	1.07

Fraser EFW subset ranked by raw importance; shares use total forest importance, not the Fraser EFW subtotal.

Source: Completed Objective 3 batch, 29 September 2026; 07_obj3_global_iv_cv_TEA. Scores and importance are taken directly from saved CSV outputs.